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Time Series Analysis of Monthly Live Births in South Korea (1997-2024)

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1. Introduction

1.1 Significance of the Study

This study aims to conduct a complete time series analysis of monthly live births in South Korea from 1997 to 2024. Is birth rate actually declining? Is there a specific month where birth rates are the highest? What would be the number of live births after five years of time?

The study aims to analyze the behavior of the data using classical and modern approaches, examine the stationarity of the series, identify suitable models using the Box–Jenkins approach, and generate forecasts for future live birth values, which will help understand the demographic changes in South Korea better and highlight the importance of forecasting methods in economic planning.

1.2 Background & Literature Review

According to the World Health Organization (WHO), a live birth is a complete extraction of a baby from its mother that then breathes or shows any sign of life. Analyzing the number of births over time and accurate forecasting helps the government measure population growth, healthcare conditions, fertility behaviour, and future labor force trends, allowing policymakers to accordingly develop effective strategies to determine the number of schools, hospitals, and other institutions needed in that country. Hence, forecasting future birth trends is important for demographic and economic planning.

According to several previous studies on fertility and birth rates in South Korea, it was found that the country has the lowest fertility rate among OECD countries with a long-term downward trend (Jung,2024), which is raising concerns about population aging, labor shortages, and economic productivity. Experts have justified that dropping fertility rates are based on rising housing costs, education, lack of jobs, and anxiety of young people about their future (Yoo, 2023), which makes it an interesting country to analyze prevailing birth trends.

2. Data Analysis

2.1 Data Set Overview

The data set used in this study consists of monthly live birth records in South Korea in the period from January 1997 to December 2024. The data set contains 336 monthly observations.

It was obtained from KOSIS (Korean Statistical Information Service), the national statistical database, operated by the Ministry of Data and Statistics in South Korea. KOSIS provides an easy access to reliable statistical information collected by government agencies.

Our variable in this project is the total monthly number of live births, measured in persons. The data set contains no missing data points, making it suitable for time series analysis.

2.2 Graphical Exploration of the Time Series

Our time series graph displays the monthly number of births in South Korea from 1997 to 2024, providing an expansive overview of general societal changes within the country.

The overall trend that follows the time series is a clear and strong downward trend, further indicated by the births dropping from an average of 60,000 per month in 1997 to 20,000 per month in 2024. An approximately 67% decline.

As clearly indicated by the graph, regular fluctuations are apparent each year, which means there is a consistent up and down spike throughout the time series.

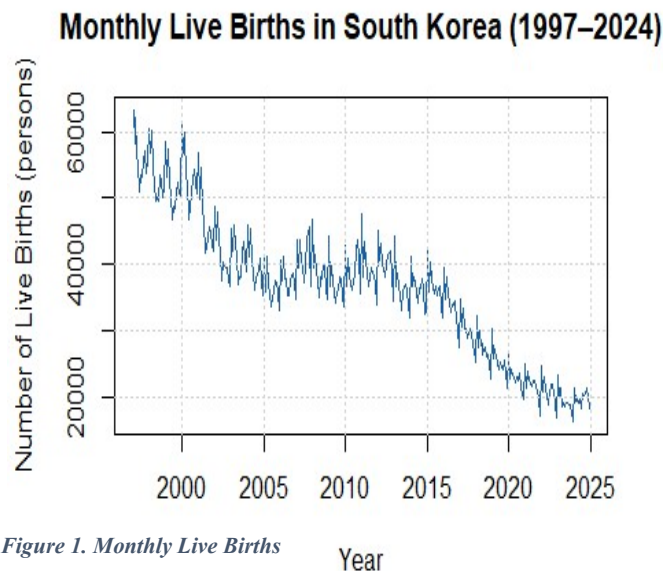


Figure 1. Monthly Live Births

From the two previous points, we could deduce a clear non-stationary time series, as both the mean and variance change over time, indicating a need to apply transformations before going through with any modelling.

2.3 Trend & Seasonal Decomposition

A multiplicative decomposition method was used for our time series, which is most appropriate as the seasonal fluctuations shrink as the overall live births decline.

As for the trend, we could observe a persistent downward movement throughout the time series that gets more accelerated by the time we hit 2015, which suggests that there is a deeper problem within South Korea that could be catastrophic for the country if it is not already too late.

For the random component, there are almost consistent fluctuations around 1, indicating a systematic pattern. Meaning the multiplicative model has mostly been able to capture both trend and seasonality.

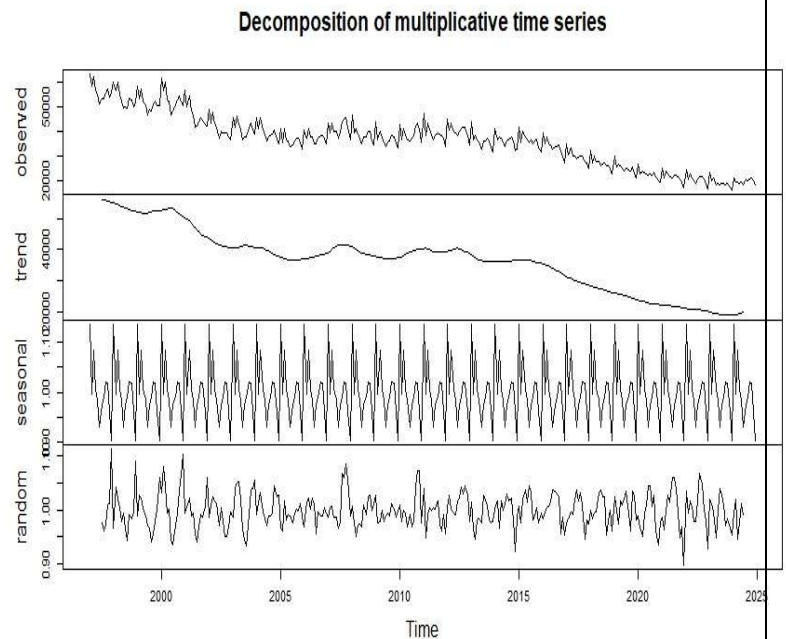


Figure 2. Decomposition of Time Series

The seasonal component almost displays a stable and consistent pattern repeating every 12 months. Birth tends to reach its highest in January (1.1350), and reaches its lowest levels in December (0.9038), as shown in the figure below.



Figure 3. Seasonal Indices

2.4 Acquiring the best trend model

What was done?

The deseasonalized birth series was regressed on three polynomial trend models, (*linear, quadratic and cubic*) to capture the underlying long-run movement after removing the seasonal effect as shown in the above section.

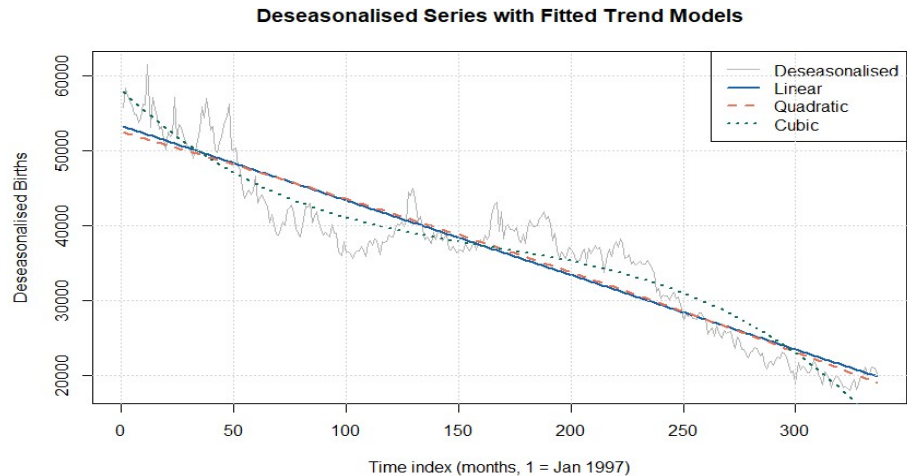


Figure 4. Deseasonalised Series & Fitted Trend Models

The model was selected based on forecast accuracy measures (*MAPE, MAD, RMSE*), rather than purely statistical measures.

A summary on the three models were as following:

- **Linear:** Straight declining trend of ~99.6 births/month, Adj. $R^2 = 0.8782$, but too rigid for a long series.
- **Quadratic:** Marginal improvement over linear (Adj. $R^2 = 0.8791$) with a barely significant squared term ($p = 0.063$).
- **Cubic:** All terms highly significant ($p < 2e-16$), capturing a non-monotonic trend with Adj. R^2 jumping to 0.9208.

An accuracy metric comparison table was provided:

<i>Model</i>	<i>MAPE</i>	<i>MAD</i>	<i>RMSE</i>	<i>MPE</i>
linear	8.40%	2,990	3,591	-1.12
Quadratic	8.17%	2,969	3,573	-1.01
Cubic	7.10%	2,358	2,887	-0.58

The Cubic model is the clear winner without a shadow of a doubt, since it wins in every metric.

2.5 Stationarity & Transformations

• Test for stationarity

A time series is stationary if its mean, variance, and autocovariance are constant over time. There are multiple ways to test for stationarity, one of which is visual inspection. The time series is non-stationary if there is an upward or downward trend, changing volatility over time, or repeated patterns.

Based on **Figure 1** and the results of the decomposition step, the series has an obvious downward trend and seasonality, which makes the series non-stationary.

Another way to test for stationarity is to apply tests; we applied the common tests for stationarity:

1. Augmented Dickey-Fuller (ADF) test.

Null hypothesis (H_0): The series is non-stationary.

Alternative hypothesis (H_1): The series is stationary.

The ADF test statistic was **-2.6703** with a p-value of **0.2939**. Since the p-value > 0.05 significance level, we fail to reject the null hypothesis, which means that the series is non-stationary at the 5% level, which was expected because we had a trend.

2. The Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test was used as a complementary test for stationarity.

Null hypothesis (H_0): The series is stationary.

Alternative hypothesis (H_1): The series is non-stationary.

The KPSS test statistic was **0.51161**, with a p-value of **0.01**. Since the p-value < 0.05 , we reject the null hypothesis, suggesting that the series is non-stationary again at the 5% level.

• Transform to a Stationary Time Series

Since the time series is non-stationary in mean (has a trend) and has seasonality, we applied the first regular difference to remove the trend, followed by the seasonal difference at lag 12 to remove the monthly seasonality.

As shown in **Figure 4**, after removing trend and seasonal effects, the series became stationary.

We reapplied the ADF and KPSS tests to confirm stationarity, and the p-value for the ADF test was 0.01

Which is lower than 0.05, indicating a stationary time series, and the KPSS yielded a p-value of 0.1

Which is greater than 0.05, indicating again a stationary time series at a significance level of 0.05.

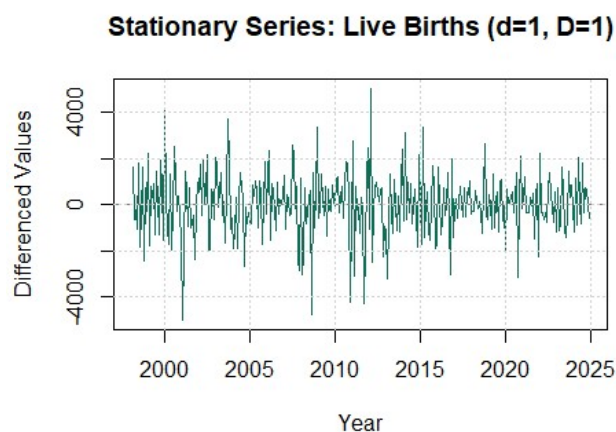


Figure 5. Series After Differencing

2.6 ACF & PACF Analysis

The ACF and PACF were computed on the stationary series (after $d=1$, $D=1$ differencing) with a maximum lag of 48 months (four seasonal cycles), as shown in **Figure 5**.

Fig 9a. ACF — Stationary Series ($d=1$, $D=1$)

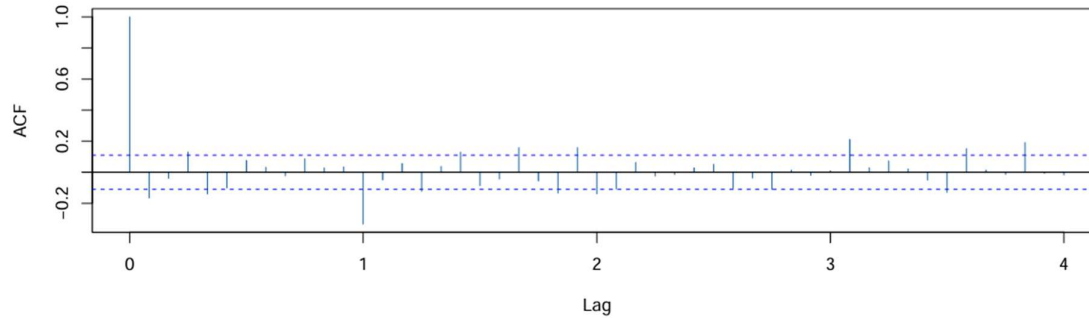


Fig 9b. PACF — Stationary Series ($d=1$, $D=1$)

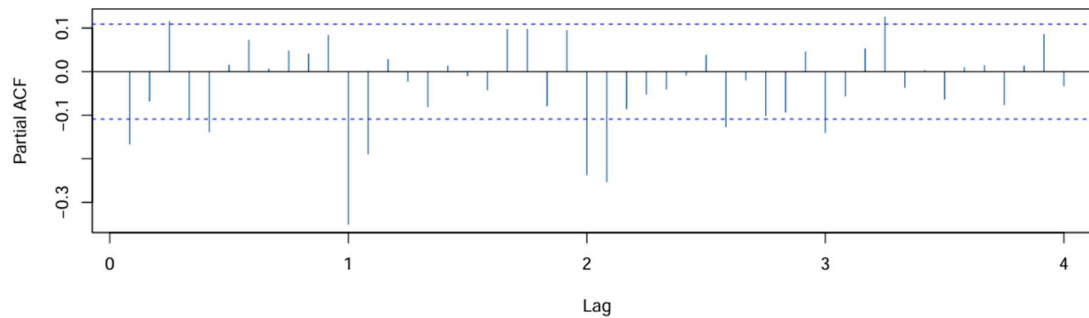


Figure 6. ACF & PACF of Stationary Series

Non-seasonal component: The ACF shows a significant negative spike at lag 1 that cuts off sharply with lag 2 being insignificant, while the PACF displays a more gradual decay. This pattern is characteristic of a moving-average process, suggesting an $MA(q)$ component. Multiple significant lags (up to lag 4) in the PACF indicate that q may be as high as 4, which is consistent with the `auto.arima` selection of $q = 4$ (discussed in the next section).

Seasonal component (lags 12, 24, 36): There is a single dominant negative spike at lag 12 in the ACF that cuts off with no significant spike at lag 24. The PACF shows a decaying pattern at the seasonal lags. This pattern is consistent with a seasonal $MA(1)$ term ($Q = 1$). No significant seasonal AR pattern is evident, so $P = 0$.

Based on this analysis, candidate models chosen included **SARIMA(0,1,1)(0,1,1)[12]**, which is the classic airline model, and **SARIMA(1,1,1)(1,1,1)[12]** as a more general alternative, in addition to the `auto.arima` search described in the following section.

2.6 ARIMA Model & Diagnostic Tests

According to Box-Jenkins methodology, we first used auto.arima with a complete search (stepwise = FALSE) across all SARIMA(p,1,q)(P,1,Q)[12] combinations to identify a good model. The search evaluated over 80 candidate models. The selected model was ARIMA(0,1,4)(0,1,1)[12], with AIC = 5488.48.

Three manual candidate models were fitted and compared as discussed earlier:

- **Model 1: SARIMA(1,1,1)(1,1,1)[12]:** AIC = 5499.55, BIC = 5518.44
- **Model 2: SARIMA(0,1,1)(0,1,1)[12]:** AIC = 5496.78, BIC = 5508.12
- **Model 3: SARIMA(1,1,0)(1,1,0)[12]:** AIC = 5542.66, BIC = 5553.99

Since the model selected by auto.arima, **ARIMA(0,1,4)(0,1,1)[12]**, achieved the lowest AIC (5488.48) compared to the candidate models, it was selected as the final model. Our model has no AR terms at either level; instead it uses four non-seasonal MA terms and one seasonal MA term to capture all the information after differencing.

The estimated coefficients are: MA(1) = -0.255, MA(2) = 0.062, MA(3) = 0.076, MA(4) = -0.194, and SMA(1) = -0.636, with $\sigma^2 = 1,346,378$.

Since **ARIMA(0,1,4)(0,1,1)[12]** contains no AR terms ($p = 0, P = 0$), there are no AR roots to examine and causality (*stationarity*) is not a concern. For invertibility, the roots of both the non-seasonal MA(4) and seasonal MA(1) polynomials were computed. All non-seasonal root moduli (1.4615, 1.5953, 1.4615, 1.5141) and the seasonal root modulus (1.5731) lie strictly outside the unit circle, confirming the model is fully invertible.

Invertibility requires that all root of MA polynomial have modulus strictly greater than 1. Which was achieved for all moduli, confirming the model is invertible.

Diagnostic tests on the final model residuals:

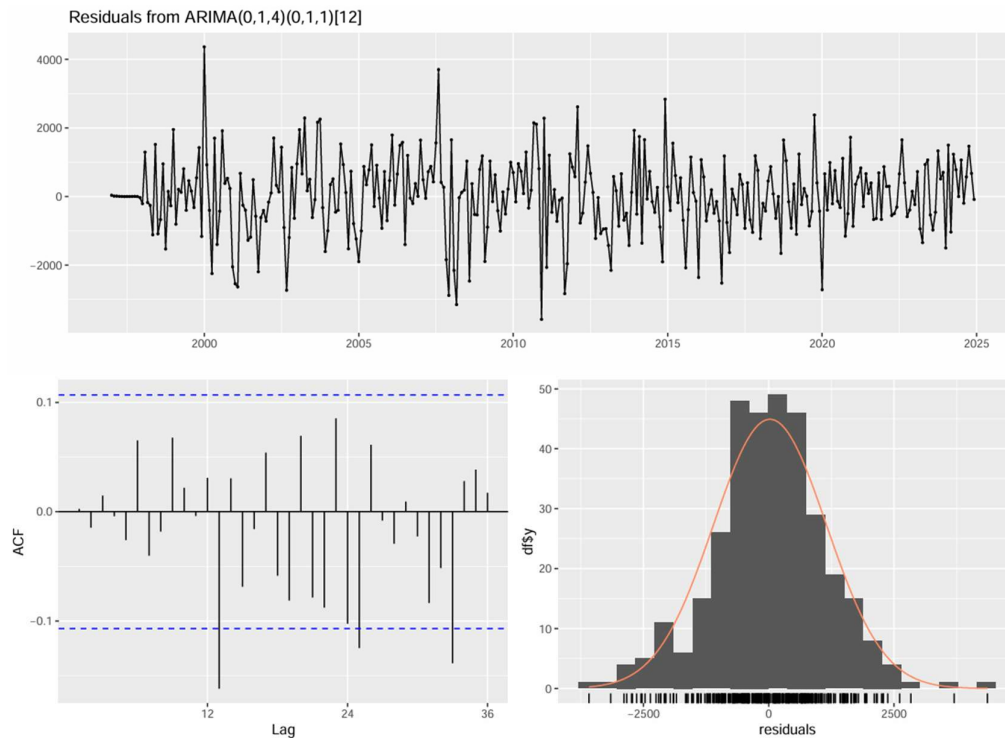


Figure 7. Residual Diagnostics of Chosen Model

Ljung-Box test:

Null hypothesis (H_0): Residuals are white noise.

Alternative hypothesis (H_1): Residuals are not white noise.

$Q = 33.719$, $df = 19$, $p\text{-value} = 0.0198$. Since $p\text{-value} < 0.05$, we reject the null hypothesis at the 5% level. And if we chose 10% significance level, we would not have enough evidence that the residuals are not white noise. While this indicates some residual autocorrelation may exist, the effect is marginal compared to the other advantages of the model, it still represents the best-fitting specification identified by the exhaustive search. The MAPE of 2.51% confirms how good the predictive accuracy of the model is.

Shapiro-Wilk normality test:

Null hypothesis (H_0): The residuals are normally distributed.

Alternative hypothesis (H_1): The residuals are not normally distributed.

W = 0.989, p-value = 0.013. The null hypothesis is rejected at the 5% level, indicating a mild departure from normality in the residuals. However, at 10% significance, the normality would not reject normality. This is not unusual for a long monthly series and means that the forecast confidence intervals should be treated as approximate rather than exact, but the model remains the best choice as explained earlier.

2.7 Forecasting

2.7.1 One-Year Forecast via Decomposition

Using the classical decomposition results, we generated a one-year forecast for January through December 2025. The cubic trend model was projected forward and then re-multiplied by the 12 monthly seasonal indices to reconstruct the final forecast.

The decomposition method predicts a total of **138,331 births** for 2025, with an average of approximately 11,528 births per month. Across the year, January records the highest forecast (15,071 births; seasonal index = 1.135) and December the lowest (8,703 births; seasonal index = 0.904).

The trend component itself declines steeply through 2025, from 13,279 in January down to 9,630 by December, reflecting the aggressive downward extrapolation of the cubic polynomial at the boundary of the sample.

Fig 11. Live Births: Historical + 2025 Decomposition Forecast

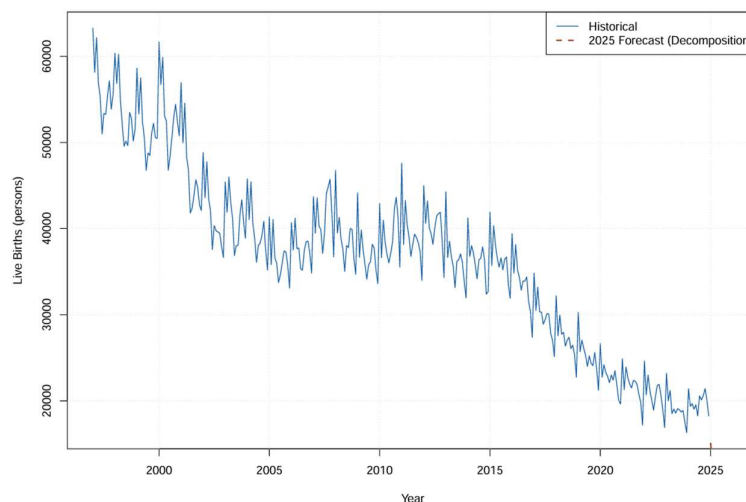


Figure 8. 2025 Forecast using Decomposition

It is important to note that this forecast suffers from polynomial over-extrapolation. The cubic trend model, while providing the best fit in sample, curves steeply downward at the end of the series. To explore this fact, the trend prediction for late 2024 was studied, and it already falls below 14,000, while the actual observed values were well above this prediction, ranging from 18,000 to 21,000 per month in that period. Consequently, it can be said that the decomposition forecast for 2025 significantly underestimates expected births relative to the ARIMA forecast, and should be treated with caution.

2.7.2 Five-Year Forecast via ARIMA

Using the final ARIMA(0,1,4)(0,1,1)[12] model, we generated a five-year forecast covering January 2025 through December 2029 (60 months), with 80% and 95% prediction intervals.

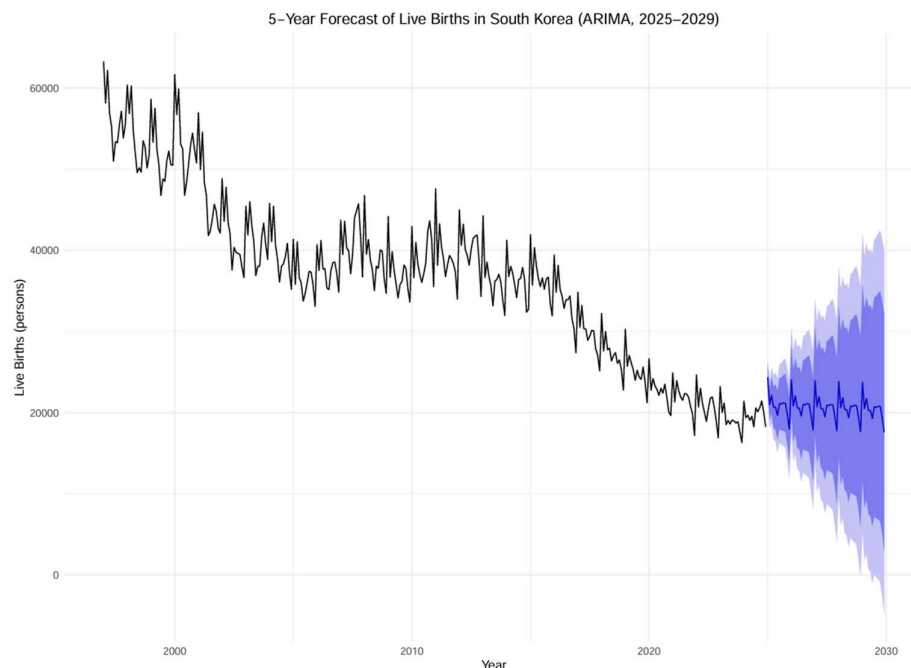


Figure 9. 2025-2029 Forecast using ARIMA

Annual forecast totals (ARIMA): 2025: 250,745 | 2026: 249,222 | 2027: 247,993 | 2028: 246,764 | 2029: 245,535.

The ARIMA forecast predicts a gradual, continued decline in annual births from approximately 250,745 in 2025 to 245,535 by 2029. Monthly values remain in the range of roughly 17,500–24,500 throughout the forecast, which is consistent with the most recent observed levels of 18,000–21,000 per month in 2024. **The ARIMA** model relates its forecast directly to the most recent observations because of the differencing mechanism, without imposing a rigid functional form on the trend. This is why it produces far more realistic forecasts than the decomposition method.

As the forecast horizon extends toward 2028–2029, we can see how the prediction intervals widen, reflecting growing uncertainty. By December 2029, the 95% interval spans from approximately -4,775 (lower bound) to 39,887 (upper bound). The negative lower bound indicates that very low birth counts remain within the range of statistically acceptable outcomes. The seasonal pattern (peaks in January and March, troughs in November and December) still exist throughout the forecast period.

Comparison of the two forecasts for 2025: The decomposition method predicts an annual total of 138,331 births (avg. 11,528/month), while ARIMA predicts 250,745 (avg. 20,895/month).

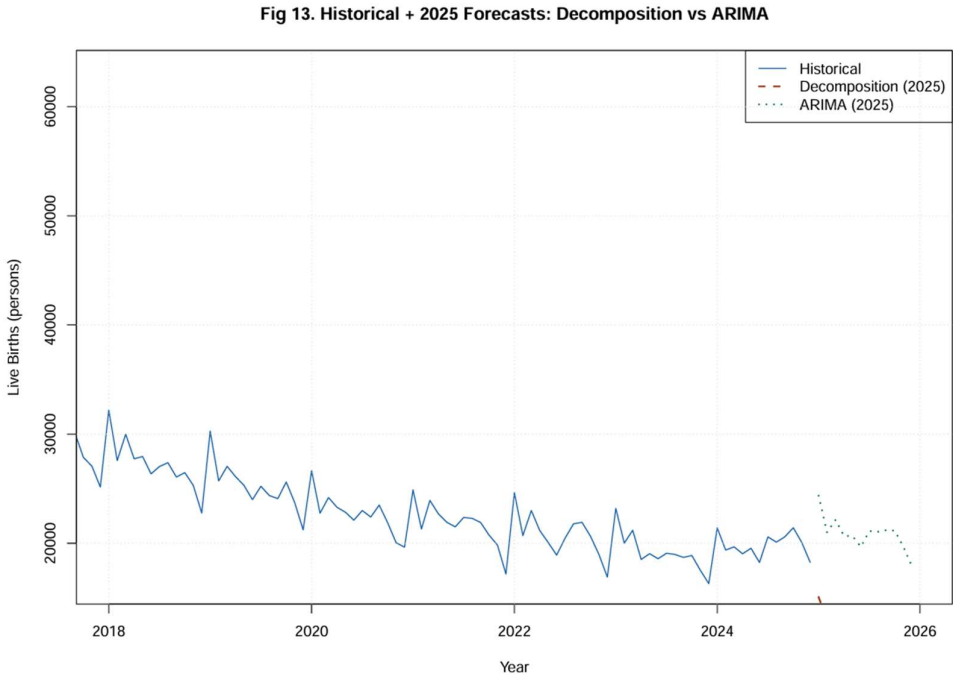


Figure 10. 2025 Forecast: Decomposition vs ARIMA

The gap of roughly 9,000 births per month is not random, as it increases steadily from mid-year through December, reflecting that the cubic trend model diverges increasingly from reality as the forecast moves further from the estimation sample. Given that actual 2024 monthly births were in the 18,000–21,000 range, the ARIMA forecast is clearly the more reliable and realistic estimate for 2025.

3. Conclusion & Real-Life Implications

The analysis that was done on South Korea's monthly live births (1997–2024) revealed a clearly accelerating and strong downward trend, with total births declining by more than 70% over the 28-year period. The classical decomposition confirmed a multiplicative seasonal pattern, with births peaking around January and diminishing in December. Among the three trend models tested, the cubic polynomial model was selected, as the *MAPE* was the lowest among the two other models (**7.10%**), best capturing the non-monotonic nature of the decline.

Stationary tests (*ADF and KPSS*) confirmed to us that the series is non-stationary, prompting us to do a combination of one regular difference ($d=1$) and one seasonal difference ($D=1$) to achieve stationarity. The ACF and PACF of the differenced series guided the identification of candidate SARIMA models, and after an exhaustive search, **ARIMA(0,1,4)(0,1,1)[12]** was selected as the final model with the lowest AIC of 5488.48. The model was confirmed to be fully invertible, and residual diagnostics indicated a good fit.

Regarding the forecasting, the decomposition method projected approximately **138,331 total births for 2025**, while the ARIMA model projected **250,745**. The large difference here is explained by the nature of the cubic trend polynomial model where it extrapolates the decline too aggressively beyond the estimation range. Thus making the ARIMA forecast more reliable.

In real life, these values carry significant implications. South Korea's continued decline in the birth rates, making it one of the lowest in any developed nations, is a sign of a deeper-rooted problem in the socioeconomic landscape and cultural pressures. Continued decline through 2025–2029 points to mounting pressures on healthcare, education, labor supply, and pension systems. Government intervention is of absolute necessity, whether that through incentives or easier passages to married life, but if left untreated the consequences would be irreversible and some might say, it is too late...

4. References

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